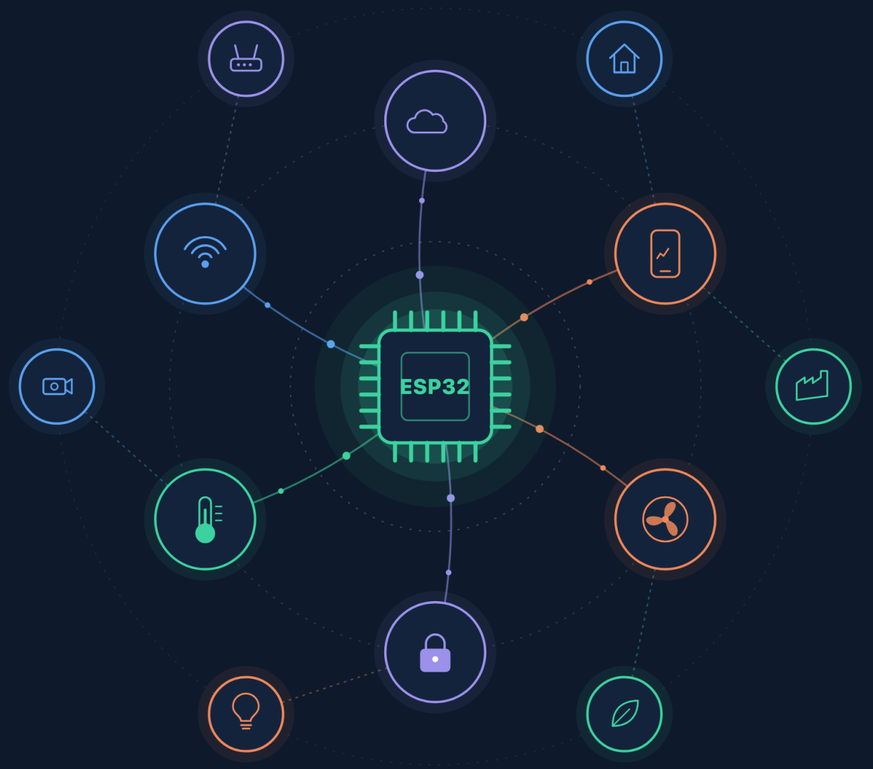
MODULE 1 · 45 MINUTES

What Is the Internet of Things?

What IoT is, what makes a device an IoT device, and how telcos already use it at tower sites and in data centres.



Mehdi Lotfinejad

Introduction to the Internet of Things

STORY ONE · THE TOWER SITE

A hilltop tower site has lost grid power. The generator is running. How much fuel is left?

The nearest engineer is hours away by road.

STORY TWO · OUR DATA CENTRE

2 a.m. In our data centre, one rack is getting hotter. Nobody is in the room.

Who finds out first: a sensor, an engineer, or a customer?

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
You are here

02
How it's built
Lecture

03
Sensors and actuators
Lab 2

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
Lab 7

HOW WE WORK TODAY

One case, two lenses

One case

Start from a real problem: the tower site on generator, or the hot rack.

Technical lens

Asks **how**: which sensor, which network, edge or cloud?

Business lens

Asks **so what**: cost, uptime, risk, payback.

At every step today, we ask **both** questions.

BY THE END OF THIS MODULE

You will be able to...

01 **Explain** what IoT is, with four words: sense, send, decide, act

02 **Spot** IoT at a tower site and in our data centres

03 **List** the characteristics that make something an IoT device

04 **Name** five ways telcos use IoT, and why each one matters

05 **Ask** two questions about any IoT idea: how, and so what?

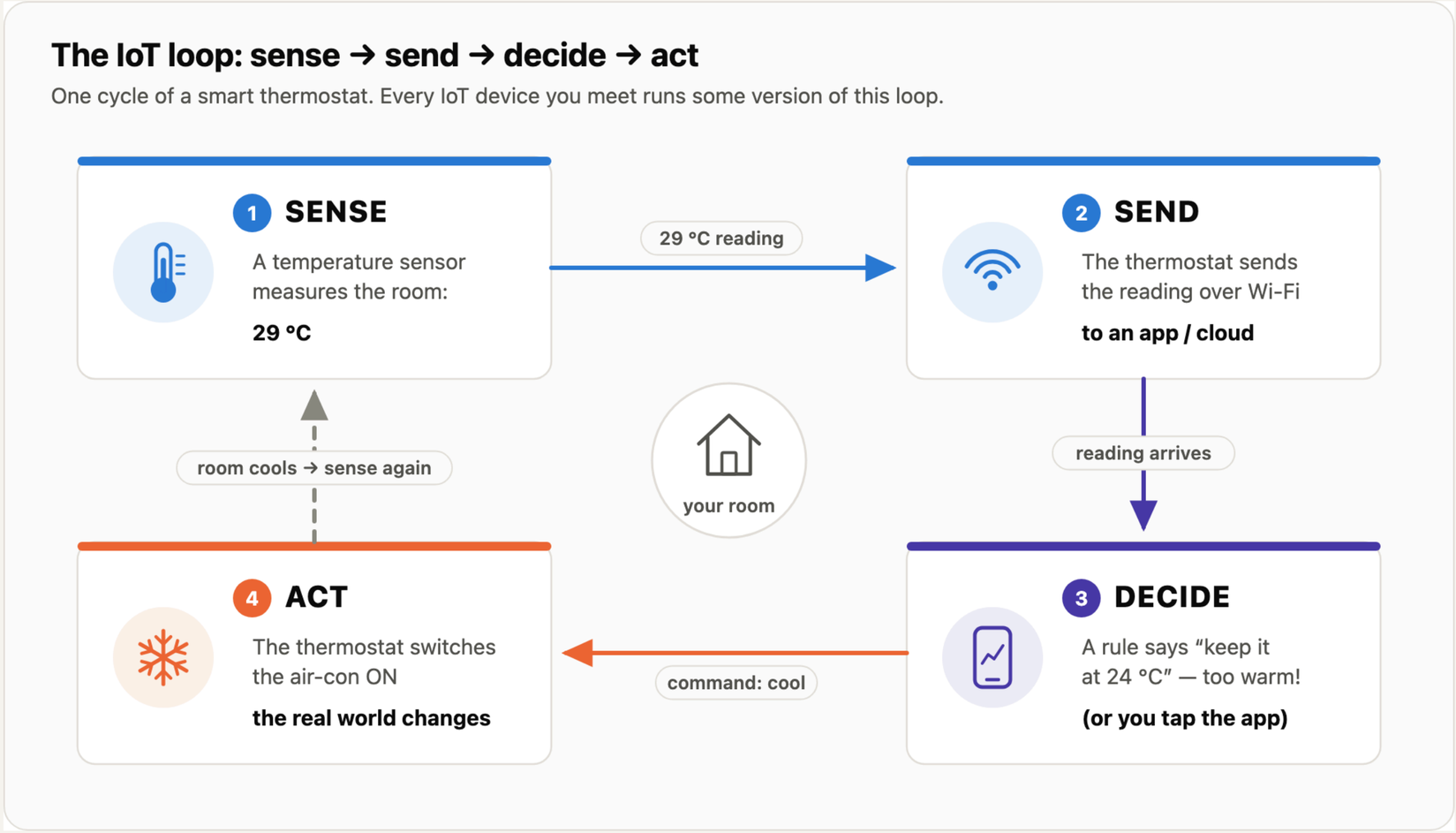
A SIMPLE DEFINITION

The Internet of Things is the network of physical objects that sense the world, share what they sense, and act on it.

A “thing” can be almost anything: a fuel tank, a shelter door, a water pump, a rack sensor, or a cow with a smart collar.

THE IOT LOOP

Sense, send, decide, act



PAUSE AND CHECK

At a tower site, a sensor measures the fuel in the generator’s tank and reports it to the NOC. When fuel is low, someone sends a truck. Can you find the four steps?

Sense → send → decide → act. Talk to your neighbour for one minute.

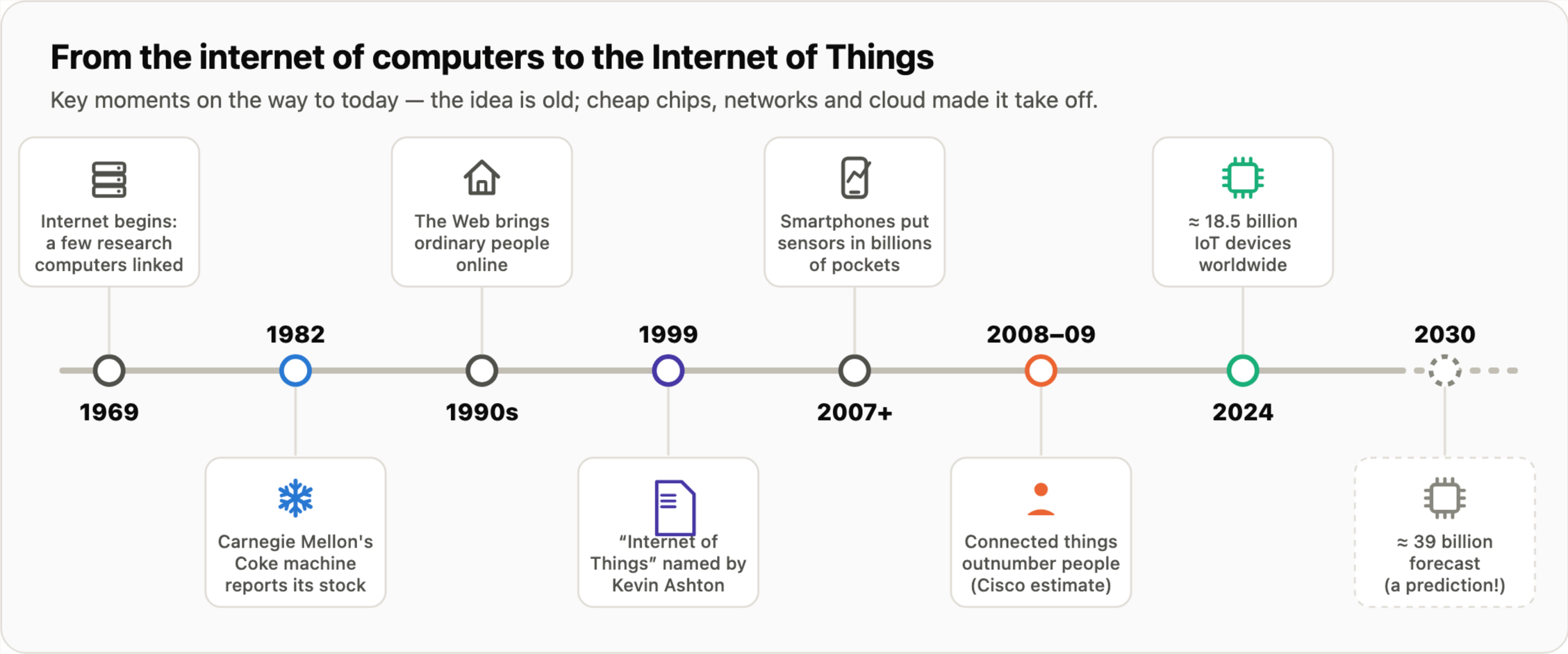
Part A

How we got here

From an internet of computers to an Internet of Things.

SIXTY YEARS IN ONE LINE

From a few computers to billions of things



WHAT MAKES A DEVICE AN IOT DEVICE?

Eight shared characteristics

Characteristic	What it means	Example
Sensing	Measures something in the physical world	A fuel sensor measures the tank
Connectivity	Sends and receives data over a network	The site reports to the NOC
Unique identity	Each device can be told apart	Each site and sensor has its own ID
Automation	Processes data and acts by itself	Extra cooling switches on at 28 °C
Ability to act	Changes the physical world	A relay starts the generator
Scale	Often huge numbers of devices	Many sites, watched from one NOC
Limited resources	Little power, memory and processing	A small sensor on a battery
Variety	Many makers, still working together	Generators and batteries from many brands

If you remember only two: it **senses or acts** on the physical world, and it is **connected**.

Part B

Why now — and why here?

The Coke machine was 1982. Why is IoT booming now, and why does it matter for a telco?

FOUR THINGS GOT CHEAP AT ONCE

Why IoT is growing so fast

Sensors

Smartphones created a huge market for tiny sensors, so prices fell for everyone.

Chips

A microcontroller with Wi-Fi built in, like the ESP32, now costs very little.

Connectivity

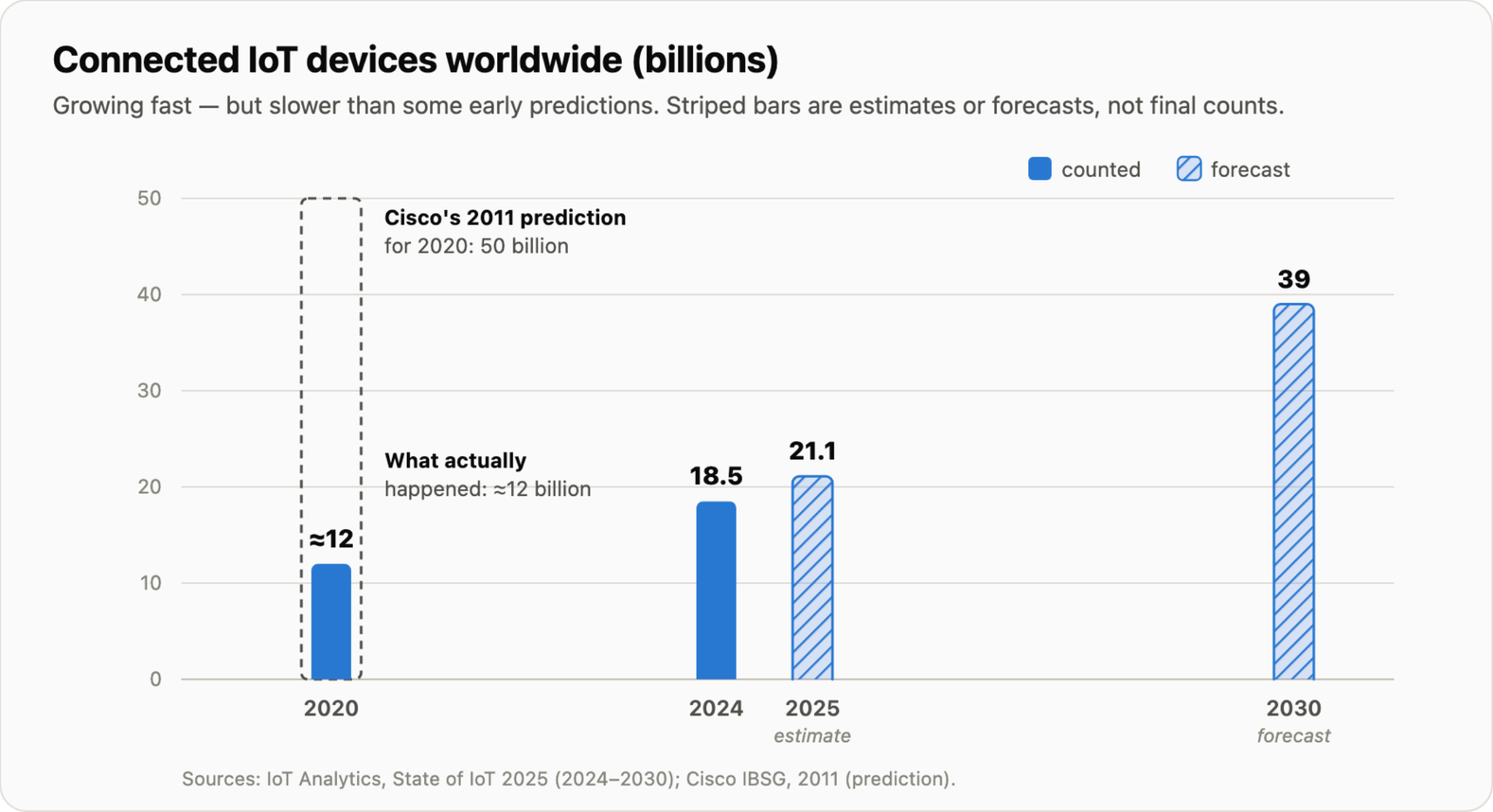
Mobile networks are almost everywhere, even at remote tower sites.

The cloud

Companies rent storage and computing by the minute, instead of buying servers.

HOW BIG IS IT?

Billions of connected devices



18.5 billion IoT devices in 2024

39 billion forecast for 2030

2020: forecast **50 billion**, real **12 billion**

Source: IoT Analytics. Dashed outline: Cisco’s 2011 forecast for 2020.

WHY THIS MATTERS HERE

Why this matters for a telco

15–20%

of a mobile operator's **operating costs** go on energy (GSMA)

1 bn

low-power cellular IoT **connections** worldwide, end of 2025

2,055

MW of data-centre capacity in Malaysia by end-2026: double

More sites, more energy, more to watch: things that **report on themselves**.

Source: GSMA Intelligence (2025); GSMA; JLL, via The Edge Malaysia (April 2026).

Part C

Where do telcos use IoT?

Five ways telcos use IoT — and our two cases, up close.

WHERE IOT IS USED

Five ways telcos use IoT

Smart tower monitoring

The NOC sees the health of every site, live.

Remote site monitoring

Fuel, battery and door, without a site visit.

Predictive maintenance

Fix batteries and generators before they fail.

Energy management

Less diesel: run the generator only when needed.

Asset tracking

Where are our generators, spares and vans?

Each one comes back **later today**.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

CASE ONE · THE TOWER SITE

IoT at the tower site

- **Sense:** fuel level, battery voltage, shelter door
- **Send:** site RMS → the NOC, by mobile network
- **Decide:** fuel falling while the generator is off?
- **Act:** send fuel or security in time
- **So what?** Fewer trips, less theft, sites on air



Photo: BalticServers.com, CC BY-SA 3.0, via Wikimedia Commons

CASE TWO · OUR DATA CENTRES

IoT in our data centres

- **Sense:** temperature, humidity, leaks, doors
- **Send:** to DCIM and the operations centre
- **Decide:** over 27 °C? (ASHRAE: 18–27 °C)
- **Act:** extra cooling on, engineer alerted
- **So what?** Cooling uses 7% to over 30% of a site’s electricity (IEA)

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ **Myth** *IoT means controlling things from your phone.*	✓ **Fact** Most IoT data never reaches a phone. Other computers use it.
✗ **Myth** *IoT is just gadgets for the home.*	✓ **Fact** Tower sites, data centres, factories and cities are a huge share.
✗ **Myth** *An IoT device must be powerful.*	✓ **Fact** Most are tiny, cheap and low-power — on purpose.
✗ **Myth** *IoT is a single technology.*	✓ **Fact** It combines sensors, chips, networks, cloud, data and security.

MIXED TEAMS OF THREE

Three roles that rotate each lab

Builder

Hands on the keyboard: pastes the code, presses Play.

Analyst

Reads the guide aloud, checks the output, writes down numbers.

Decision owner

Answers the manager task: which alerts and actions matter?

Roles **rotate** each lab, so every manager builds at least once.

TEAM ACTIVITY · 5 MIN

Where is IoT already working in your site?

1. Form a team of three: mix technical and business people.
2. Choose your first roles: builder, analyst, decision owner.
3. Discuss: where is IoT already working at your sites today?
4. Pick one example and find the loop: sense → send → decide → act.
5. Decision owner: add one business question (cost, uptime or risk).

IN TELCO OPERATIONS

What IoT changes at our sites

Today	With IoT	The benefit
Engineers drive out to check each site	The site reports fuel, battery and door	Fewer site visits
Fuel trucks run on a fixed schedule	A fuel sensor tells the NOC the level	Fewer trips; no site runs dry
Fuel theft is found days later	Alarm: fuel drops, generator off	Less theft, faster response
The NOC sees an alarm, not the cause	The site sends the readings with the alarm	Faster, better-aimed repairs

Every module asks this question: what changes in our **daily work**?

KEY TAKEAWAYS

What to remember

- IoT = physical objects that **sense, share and act** on data
- Every IoT system follows the loop: **sense → send → decide → act**
- Telcos use IoT for **towers, sites, maintenance, energy and assets**
- Our two cases all day: **the tower site**, then **our data centres**
- Two lenses: **how** does it work? **So what** for cost, uptime and risk?
- Mixed teams of three; **roles rotate** each lab

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 One shelter thermometer shows the temperature on its own screen. Another sends it to the NOC. Which one is IoT — and why?

Q2 Apply the four steps to the generator’s fuel tank at a tower site.

Q3 Name two of the five ways telcos use IoT, and one benefit of each.

Q4 Which lens? (a) Which sensor? (b) What does an outage cost? (c) Edge or cloud?

UP NEXT · MODULE 2

How an IoT System Is Built

We open the “black box”: the five layers of every IoT system, and the journey of one reading from a shelter sensor to the NOC.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop